

DEMO TODAY	BENCH EMITTER	+ MEASUREMENT	+ FIELD RIG	ALL TIERS
₹0	₹16,300–₹29,400	₹19,000–₹27,000	₹55,200–₹1,44,200	₹90,500–₹2,00,600

TIER 0	Present today	Laptop and a browser. Decision layer, real footage, outcome model.			₹0
ITEM	WHAT MATTERS	QTY	UNIT	LINE TOTAL	
Laptop + browser	opens simulator.html, no server needed	1	included	included	
Python 3.10+ and repo	optional: proves the browser isn't faking it	1	included	included	

TIER 1	Bench emitter	Produces a genuine 22-27 kHz carrier. Claims no decibels.			₹16,300–₹29,400
ITEM	WHAT MATTERS	QTY	UNIT	LINE TOTAL	
USB audio interface	192 kHz sample rate (not 48 kHz)	1	₹9,000–₹14,000	₹9,000–₹14,000	
Piezo horn tweeter	KSN1005A class, quoted 2.2k-27 kHz	4	₹400–₹900	₹1,600–₹3,600	
Class-AB amplifier module	real response at 27 kHz	1	₹800–₹2,000	₹800–₹2,000	
Heterodyne bat detector	20-40 kHz; makes it audible to the room	1	₹3,000–₹6,000	₹3,000–₹6,000	
ESP32 + relay module	hardware watchdog, hazard H23	1	₹700–₹1,500	₹700–₹1,500	
E-stop mushroom switch	de-energises the amplifier	1	₹300–₹800	₹300–₹800	
12 V PSU	bench supply for the amp	1	₹600–₹1,200	₹600–₹1,200	
Resistors, wiring, enclosure	series R: piezos are a capacitive load	1	₹300	₹300	

TIER 2	Measurement	Where predictions become measurements. Highest value per rupee.			₹19,000–₹27,000
ITEM	WHAT MATTERS	QTY	UNIT	LINE TOTAL	
Dodotronic UltraMic 192K EVO	EUR 180 ex-works; landed incl. duty	1	₹17,000–₹22,000	₹17,000–₹22,000	
Tripod + boom arm	repeatable mic placement	1	₹1,500–₹3,000	₹1,500–₹3,000	
Laser distance meter	SPL-vs-distance needs real distances	1	₹500–₹2,000	₹500–₹2,000	
Audacity / REW	free analysis software	1	included	included	

TIER 3	Field prototype	Nothing here runs near an animal without ethics approval.			₹55,200–₹1,44,200
ITEM	WHAT MATTERS	QTY	UNIT	LINE TOTAL	
Directional horn / tweeter array	the +12 dB the link budget assumes	1	₹8,000–₹25,000	₹8,000–₹25,000	
Fixed IP66 camera	1080p, must not move	1	₹3,000–₹8,000	₹3,000–₹8,000	
Rigid pole mount	calibration is void if the mount sways	1	₹3,000–₹6,000	₹3,000–₹6,000	
Edge compute	Pi 5 + Hailo-8L, or Jetson Orin Nano	1	₹18,000–₹45,000	₹18,000–₹45,000	
Solar + MPPT + battery	event-triggered duty cycle	1	₹12,000–₹30,000	₹12,000–₹30,000	
Enclosure, cable, surge protection	monsoon	1	₹6,000–₹15,000	₹6,000–₹15,000	
Calibration checkerboard	printed; distances are fiction without it	1	₹200	₹200	
Contingency	there is always contingency	1	₹5,000–₹15,000	₹5,000–₹15,000	

EVERYTHING, TIER 0 THROUGH 3	₹90,500–₹2,00,600
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Not purchasable, and required before any animal is exposed

• Animal ethics approval (IAEC)	weeks of lead time	blocks all animal exposure
• Veterinary supervision	per trial	required for formal trials
• Road authority permission	per site	required near any carriageway

Why a cheap part is the right part

Buying decisions checked against the project's own acoustics model.

The emitter: piezo horn tweeter, KSN1005A class

Frequency response	4 kHz – 27 kHz	covers 22–27 kHz. Does NOT reach 30 kHz.
Sensitivity	94 dB SPL @ 1 W / 1 m	drives the watt column below
Power handling	50 W RMS	maxes out at 111 dB @ 1 m
Unit price	₹400–900	an array of four is affordable

LINK BUDGET AT 22 kHz, +12 dB HORN, ON THIS EXACT PART			
TARGET	NEED @ 1 m	WATTS	HEADROOM UNDER CEILING
10 m	84.5 dB	0.11 W	25.5 dB
15 m	90.0 dB	0.40 W	20.0 dB
25 m	98.3 dB	2.70 W	11.7 dB
40 m	108.2 dB	26.60 W	1.8 dB

A single commodity tweeter on half a watt covers the realistic engagement range.

SAFETY PROPERTY YOU GET FOR FREE

Reaching the 142 dB the prototype paper reported would need 63,000 W from this tweeter — 1,262× its 50 W rating. The hardware physically cannot produce the exposure the governor refuses. Software says no; hardware could not comply anyway.

Three traps before you spend anything

- 1. 40 kHz modules are the WRONG part**

HC-SR04 and every hobby ultrasonic module is a narrowband 40 kHz resonator. Cattle hearing ends at 35 kHz, so it is inaudible to the animal and useless. Buy wideband piezo horn tweeters quoted to 27 kHz.
- 2. Most amplifiers stop where we start**

Audio amps target a 20 kHz ceiling and many class-D modules roll off hard just above it. Check the datasheet at 27 kHz.
- 3. A phone cannot measure this**

Phone mics are low-passed below 20 kHz. Any decibel figure quoted without a calibrated ultrasonic microphone is fabricated.

Recommendation for a two-day deadline

Do not buy the emitter. You cannot validate it in two days, and an unmeasured transducer converts your strongest asset — the honesty of the evidence envelope — into your weakest prop. If you want something physical, buy the bat detector (₹3,000–6,000, same-day in most metros): it makes the inaudible audible to the room, and every word you say about it is true. Present this sheet as the roadmap slide.